

Zoom chat for Q&A: <https://tinyurl.com/140zoom>

Lecture 1: Experimental Methods & Neuroanatomy

PSYCH/COGS 140M 2026.04.02



reminder: red = revised

- i will edit slides, sometimes up to the morning of class
 - mostly fixing typos, formatting, etc.
- when i make a larger change (e.g. more than just a typo), i'll add a red rectangle  to the top-right corner of the slide. like the one *up there* .
- the **final** updated version will be uploaded after class—within 24 hours, usually before that.

Historical Traditions



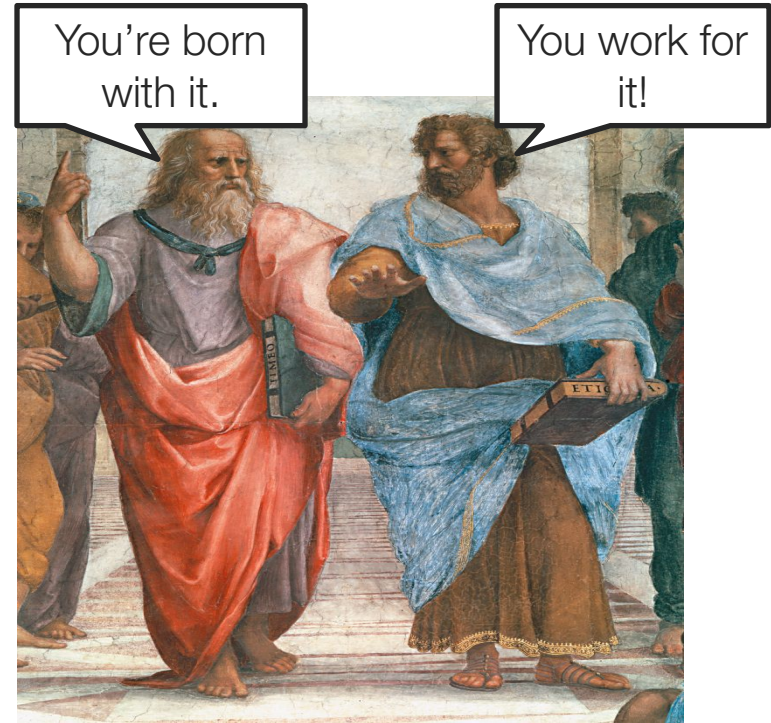
Mnemosyne
greek goddess of memory

Outline

- Historical traditions
 - Birth of experimental psychology
 - Behaviorism
 - Cognitive revolution
- Types of experimental evidence
 - Behavior
 - Neuropsychology
 - Neurophysiology
- Neuroanatomy primer

Philosophical approach

- For centuries, the study of knowledge was largely confined to philosophy
 - Nativism (nature)
 - Empiricism (nurture)



Plato

Aristotle

Herman Ebbinghaus

- Studied memory in a distinctly *scientific way*
- His approach was:
 - Experimental
 - Manipulated an **independent variable** and observed the effect on a **dependent variable**
 - *Empiricism*, distinct from *debate/introspection*
 - Quantitative
 - Expressed observations numerically



1850 – 1909

Ebbinghaus' experiments

- Performed experiments on himself
- Studied lists of nonsense words
 - WUX, CAZ, BIJ, ZOL, BAP, KEP, DAK...
- The first *word list* experiments, a general idea still used today
- Manipulated learning and retrieval variables, and observed the outcome on memory
 - Number of study trials, length of delay, etc

Ebbinghaus' key findings

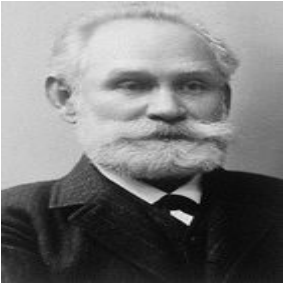
- Forgetting function is non-linear

Ebbinghaus's Forgetting Curve



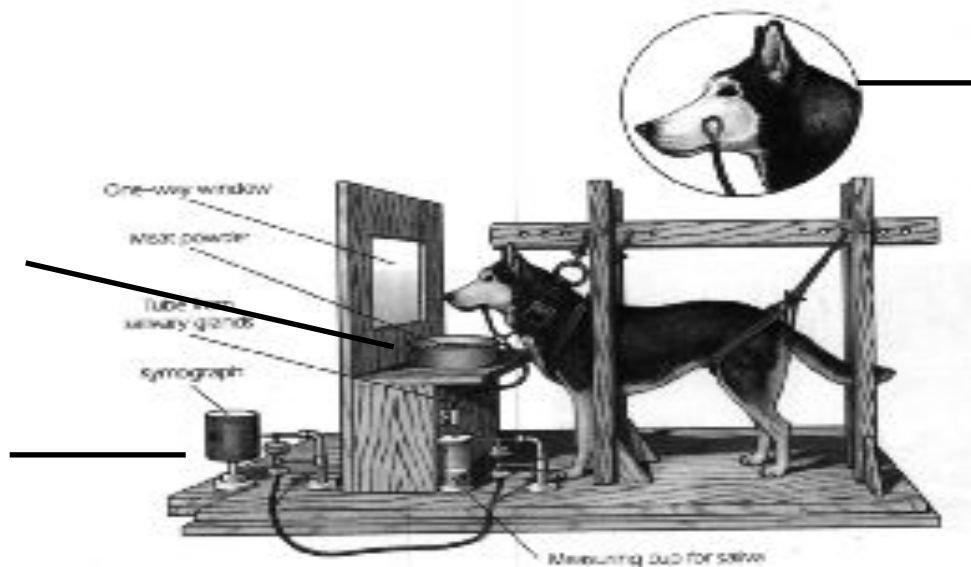
- Repetition improves memory
- Distributed practice (“spacing”/ “interleaving”) is better than massed practice (“blocking”)

Ivan Pavlov (1849 – 1936)



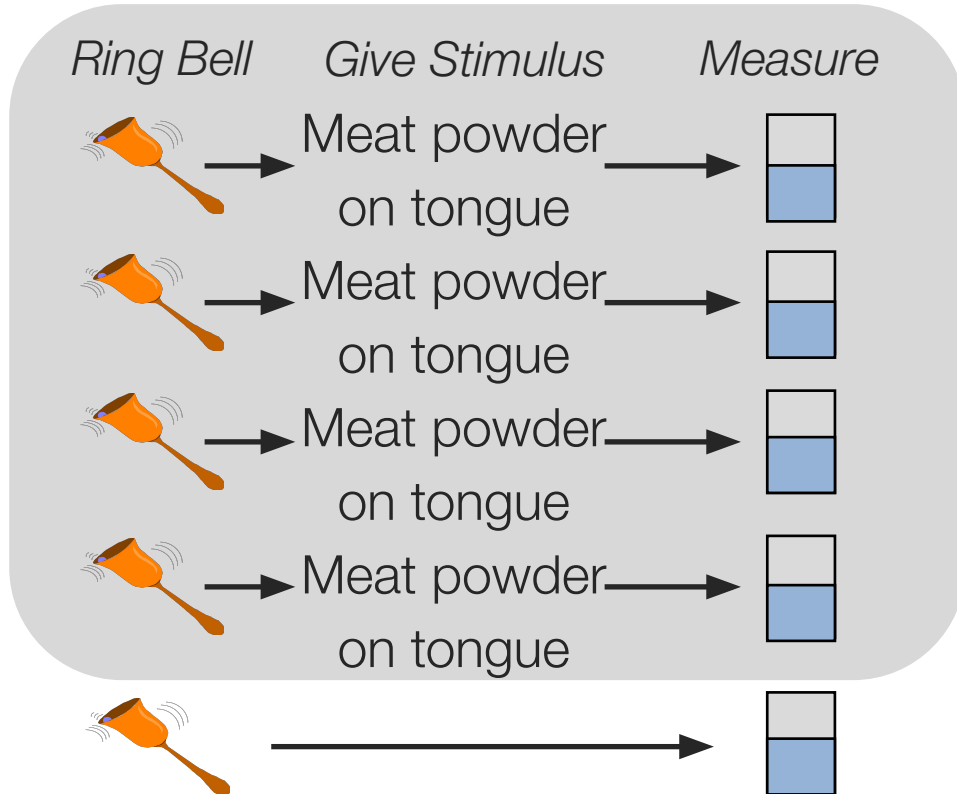
Food Dish

Recording
Device

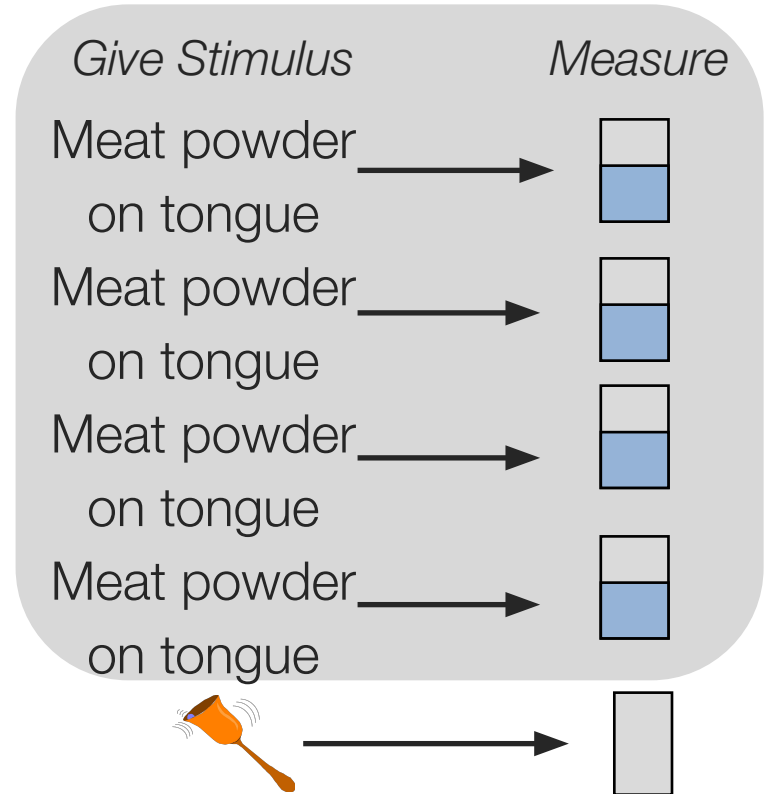


Pavlovian conditioning

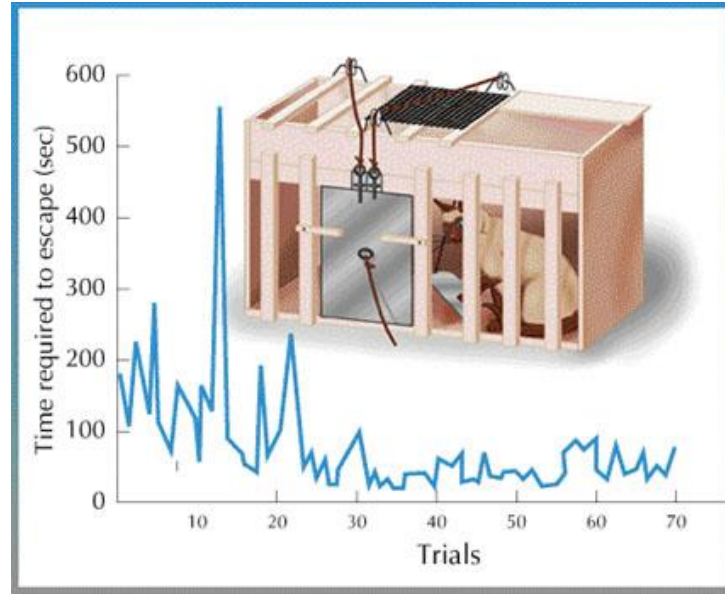
Experimental Condition



Control Condition

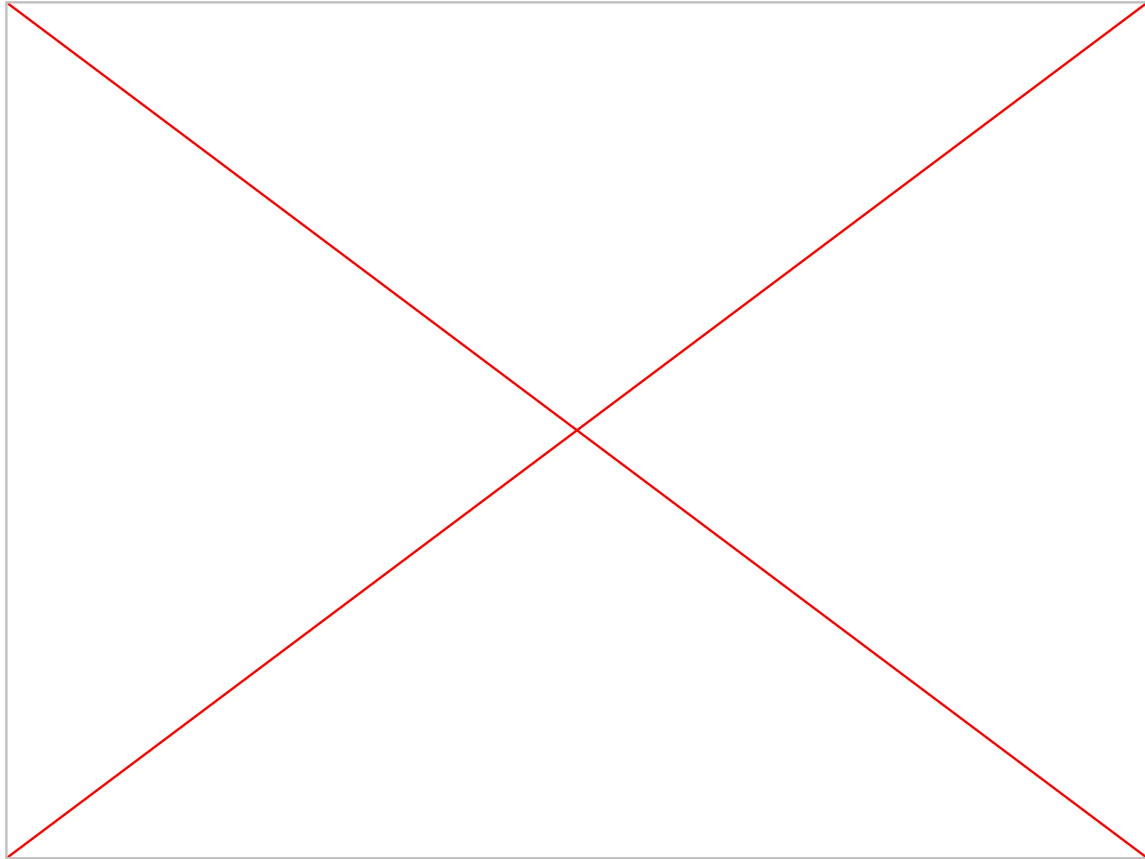


Edward Thorndike (1874 – 1949)



Law of effect: probability of a behavior changes depending on its consequences

Edward Thorndike (1874 – 1949)

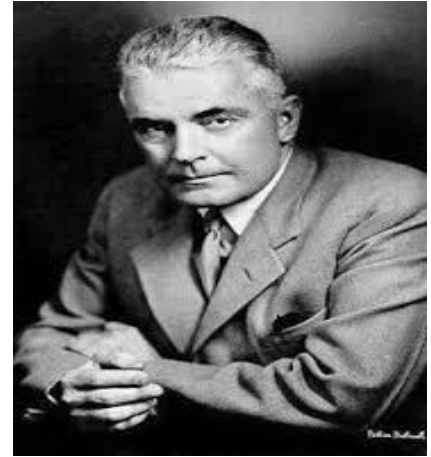


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 - **Behaviorism**
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Emergence of behaviorism

- Behaviorists thought psychology was too *subjective* and should be purely *objective*
 - Were inspired by new experimental studies (Pavlov and Thorndike)
 - Wanted to move away from internal mental processes (Freud, James) and focus on **observable behaviors**



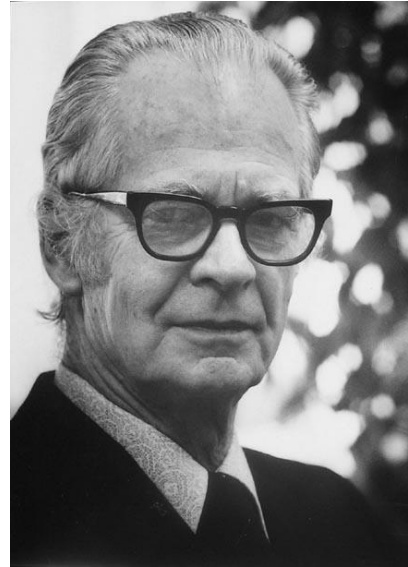
John Watson
(1878 – 1958)

Principles of behaviorism

- Focus on measurable *behaviors* (e.g. Response times, Recall accuracy, navigation success)
- Empiricism
 - Behavior can be measured and systematically explored by experimental manipulation
- Evolutionary perspective
 - Humans are similar to other animals; can learn from rodents and apply to humans
- Aimed to identify laws that could generalize to many behaviors
 - e.g. the “law of effect”

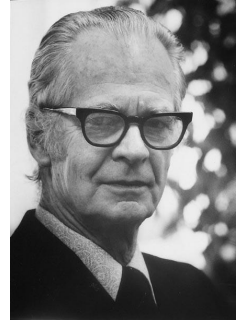
BF Skinner (1904 – 1990)

- Advocated for *radical behaviorism*
 - Free will is an illusion; behaviors are simply learned responses to stimuli
 - We need to study the environment, not the mind
 - “When what a person does [is] attributed to what is going on inside him, investigation is brought to an end. Why explain the explanation? For twenty five hundred years people have been preoccupied with feelings and mental life, but only recently has any interest been shown in a more precise analysis of the role of the environment. Ignorance of that role led in the first place to *mental fictions*...”

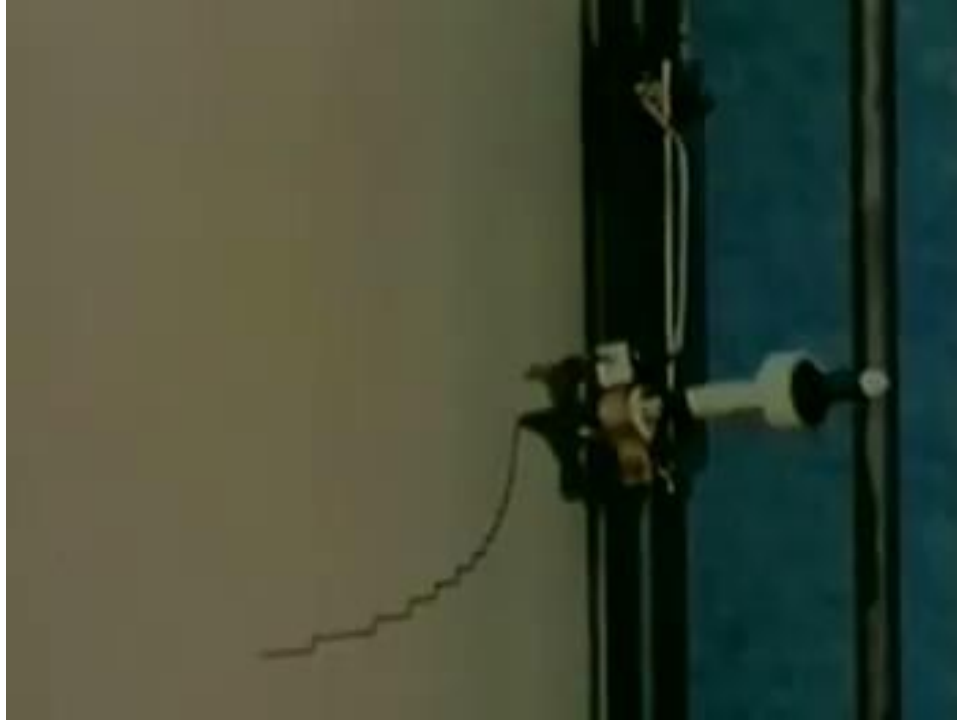


BF Skinner (1904 – 1990)

- Following Thorndike's approach, he developed the *Skinner box*
 - Automated learning device in which animals are reinforced after performing a behavior
 - e.g., press lever, receive food
 - General procedure now known as **operant conditioning** (more in Lecture 8)

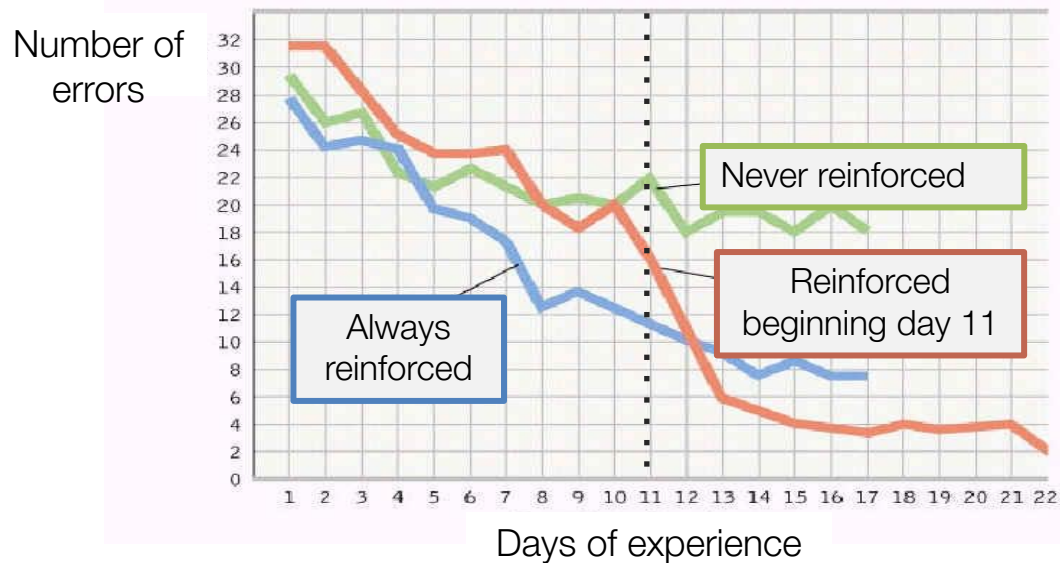


BF Skinner (1904 – 1990)



Fall of behaviorism

- *Behaviorism* had difficulty explaining all *behaviors*
- Edward Tolman (1886 – 1959)



Latent learning

Learning can occur in the absence of reinforcement and without an observable change in behavior

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Cognitive approach (1950s – present)

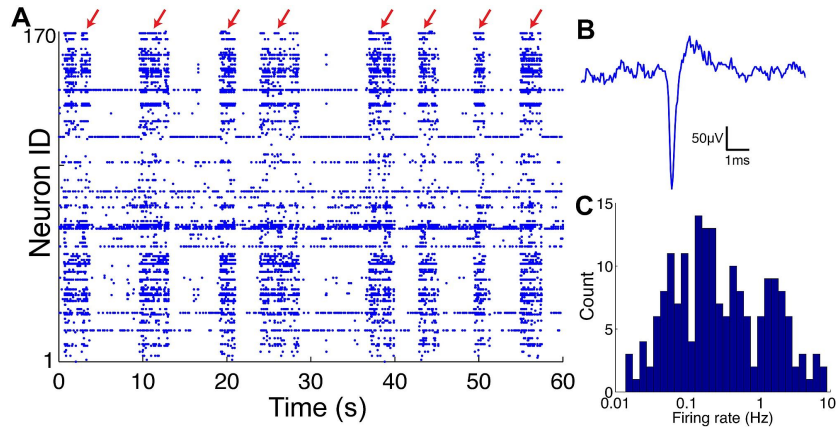
- Adopted the empirical aspects of behaviorism, but expanded focus to study *internal* processes
 - Brain/computer metaphor
 - Mathematical models and simulation
 - Humans are similar to other animals, but also have unique specializations (e.g., language, reasoning, planning)
 - Tended to neglect neuroscience (mainly due to methodological limitations)

Cognitive neuroscience (1990 – present)

- Interdisciplinary; merging of cognitive psychology and neuroscience
- Developed from several avenues of research:
 - Lesion studies (neuropsychological evidence) showing dissociations between cognitive tasks
 - Neurophysiological studies in animals
 - Development of neurocomputational models
 - Development of human neuroimaging techniques

Experimental Methods:

Types of Experimental Evidence



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Measuring Behavior



Metrics

1. **Accuracy**
2. **Latency (how fast)**
3. **Order**

Recall: What did you encode?
Recognition: Did you see *this* before?

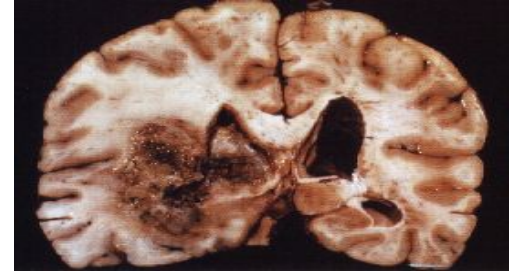
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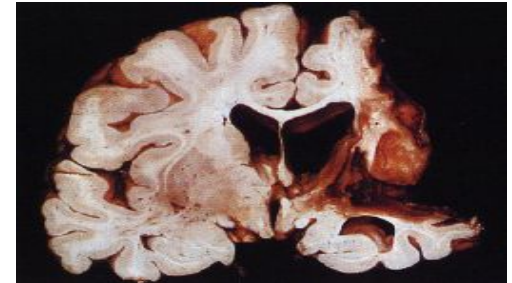
Neuropsychological evidence

- How does **damage, injury, or illness** of specific brain regions influence cognition and behavior
- *Structure – Function relationship*
- Some examples of causes of brain damage/injury
 - Tumors
 - Stroke
 - Head trauma
 - Apoxia (lack of oxygen)
 - Degenerative disease

Tumor



Stroke



Neuropsychological evidence

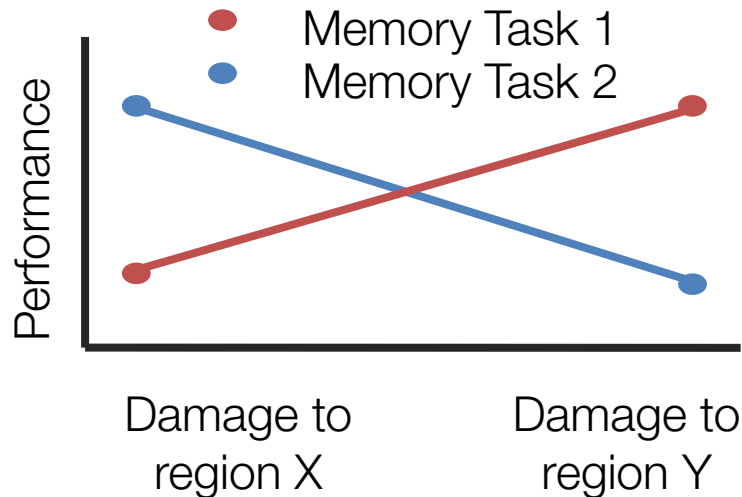
- Suzanne Corkin studied patient H.M. for over *4 decades*
- Her work yielded fundamental insight into the necessity of the hippocampus for episodic memory
- Also identified important ***dissociations:***
 - e.g., hippocampal damage impairs ‘memory process X’, but not ‘memory process Y’



Suzanne Corkin
(1937 – 2016)

Dissociations

- Lesion to region X affects task 1, but not task 2 (single dissociation)
- Lesion to region Y affects task 2, but not task 1 (single dissociation)
- If both of the above are true, that's a double dissociation!



Inference: Tasks 1 and 2 rely on different brain regions

Pros/cons of neuropsychology studies

- Pros
 - Reveals the necessity of a brain region for a given memory process
 - I.e., tells us about **causation**
- Cons
 - Damage is often not specific to one area
 - Reorganization of the brain can occur after injury, allowing for compensation
 - Unclear which stage of memory is affected

Outline

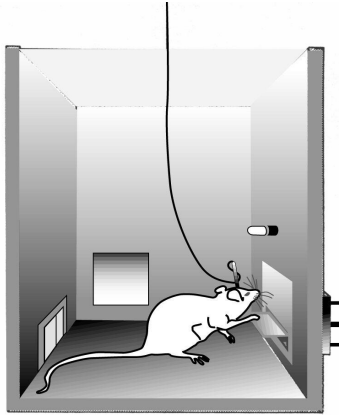
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Neurophysiology

Generally...

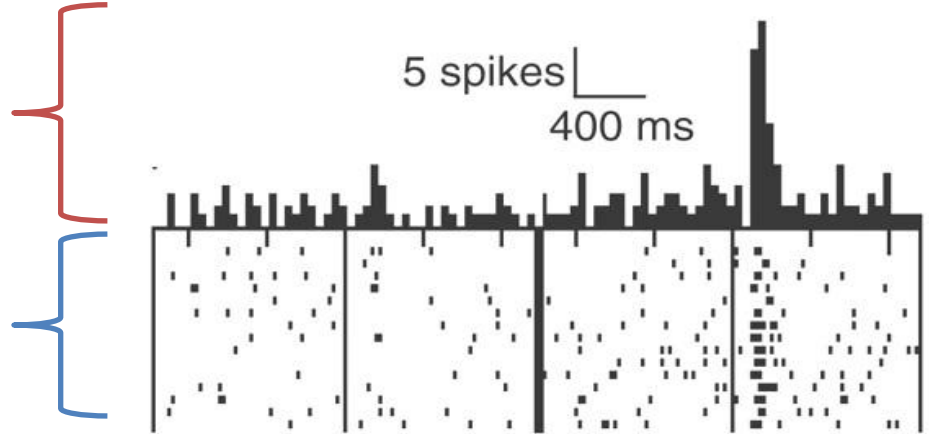
- Non-human animal studies
 - Direct recordings from single neurons
- Human studies
 - Non-invasive neuroimaging

Non-human animal recordings



Spike frequency
histogram

Raster
plot



Non-human animal recordings

- Pros
 - Direct recordings from neurons allows for excellent spatial and temporal resolution
- Cons
 - Can't tell you about causation, only **correlation**
 - Limited coverage of the brain
 - If recording from non-human animals, it is difficult to study the neural basis of higher-order cognition, e.g., language

Human recordings

- Two types of neuroimaging
 - Structural
 - Images the structure and composition of brain tissue
 - E.g., CT scan, magnetic resonance imaging (MRI)
 - Functional
 - Images brain function during a particular task

Functional neuroimaging

Electrophysiological

- Measure electrical signals on the scalp related to neural activity
 - Electroencephalography (EEG)
- Good temporal resolution; poor spatial resolution

Hemodynamic

- Measure changes in metabolism or blood flow correlated with neural activity
 - Positron emission tomography (PET)
 - Functional MRI (fMRI)
- Poor temporal resolution; good spatial resolution

Functional neuroimaging

- Pros

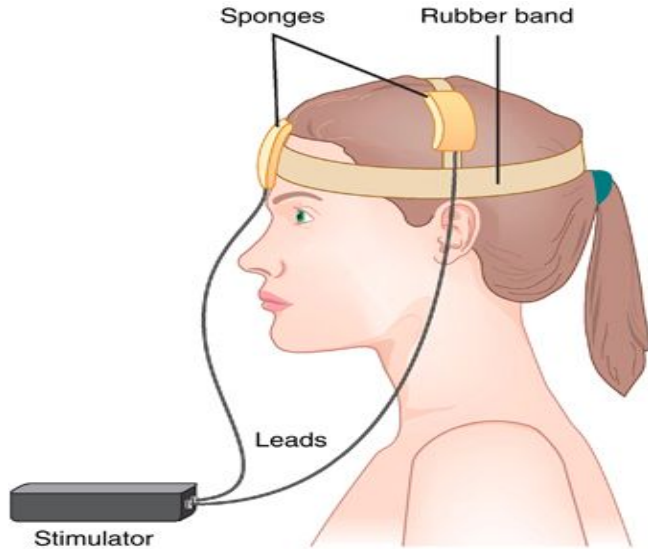
- Non-invasive
- Can evaluate function in multiple regions simultaneously
- Can be measured *before* memory is expressed (during encoding or delay)

- Cons

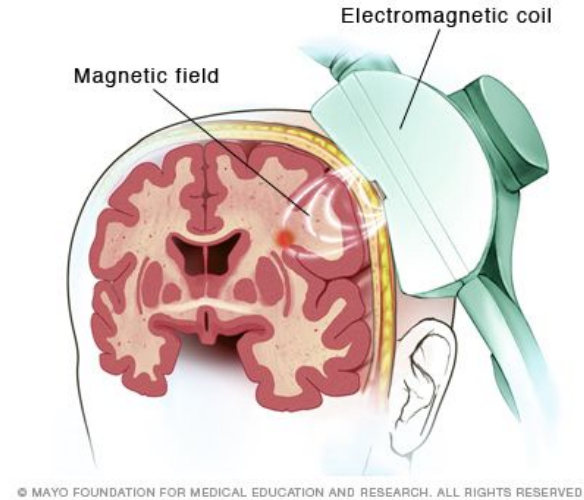
- Can't tell you about causation, only **correlation**
- Relative to direct animal recordings, poor spatial resolution

Neurostimulation

tDCS: Transcranial
direct-current stimulation



TMS: Transcranial
magnetic stimulation



Neurostimulation

- Pros

- Non-invasive (relatively)
- Can measure *a form of* causation
- Can be applied interleaved with cognitive function, within-subject

- Cons

- Unclear relevance to *in-vivo* neural mechanism (**big caveat**)
- Relatively poor spatial resolution

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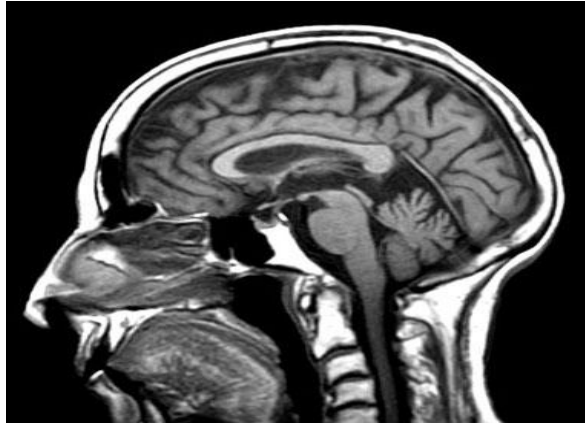
Experimental Methods: Neuroanatomy Primer



Outline

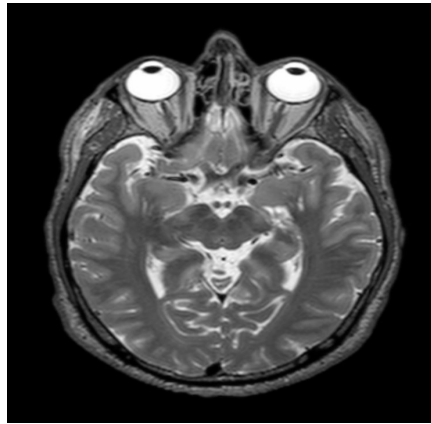
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Brain sections



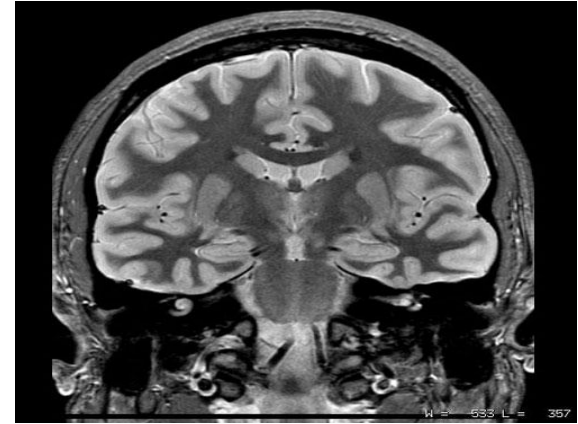
Sagittal

(Brain from the side)



Axial

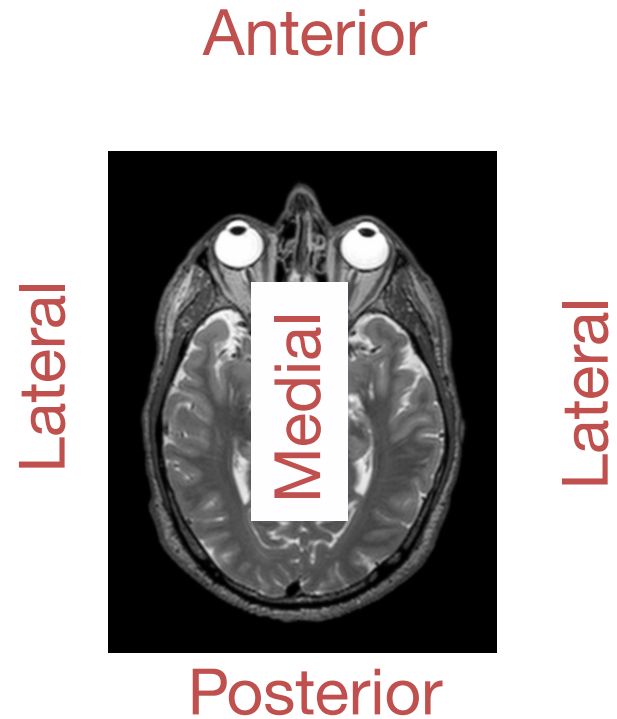
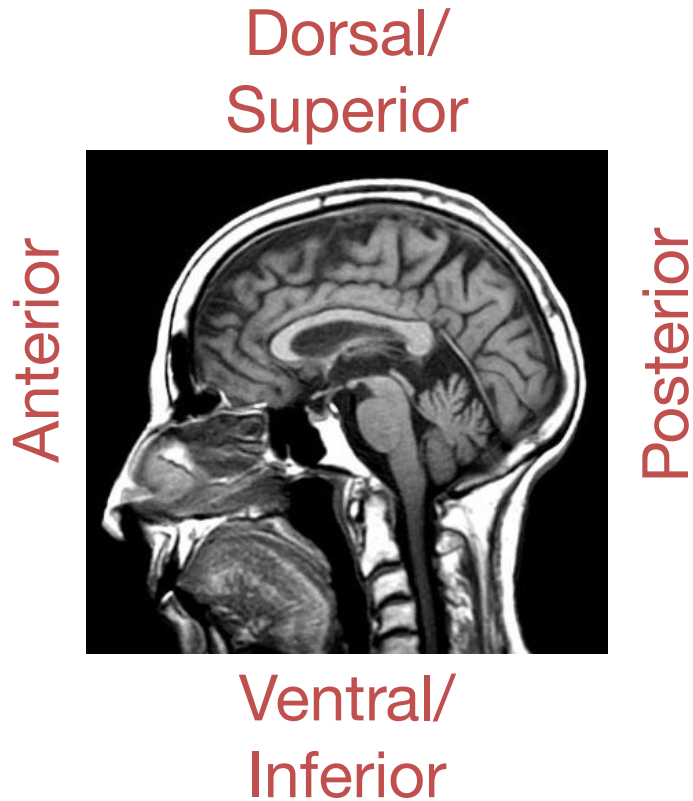
(Brain from above)



Coronal

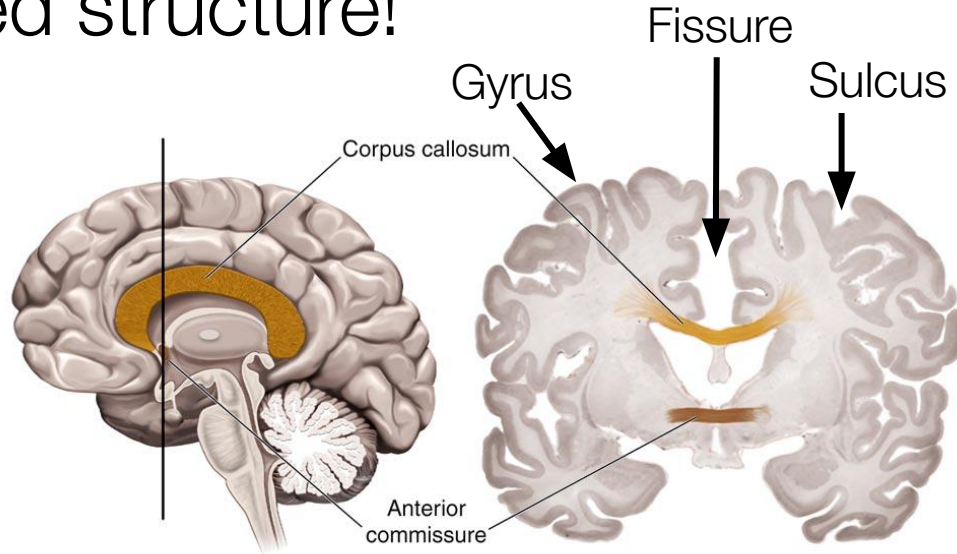
(Brain from the front)

Directional terminology

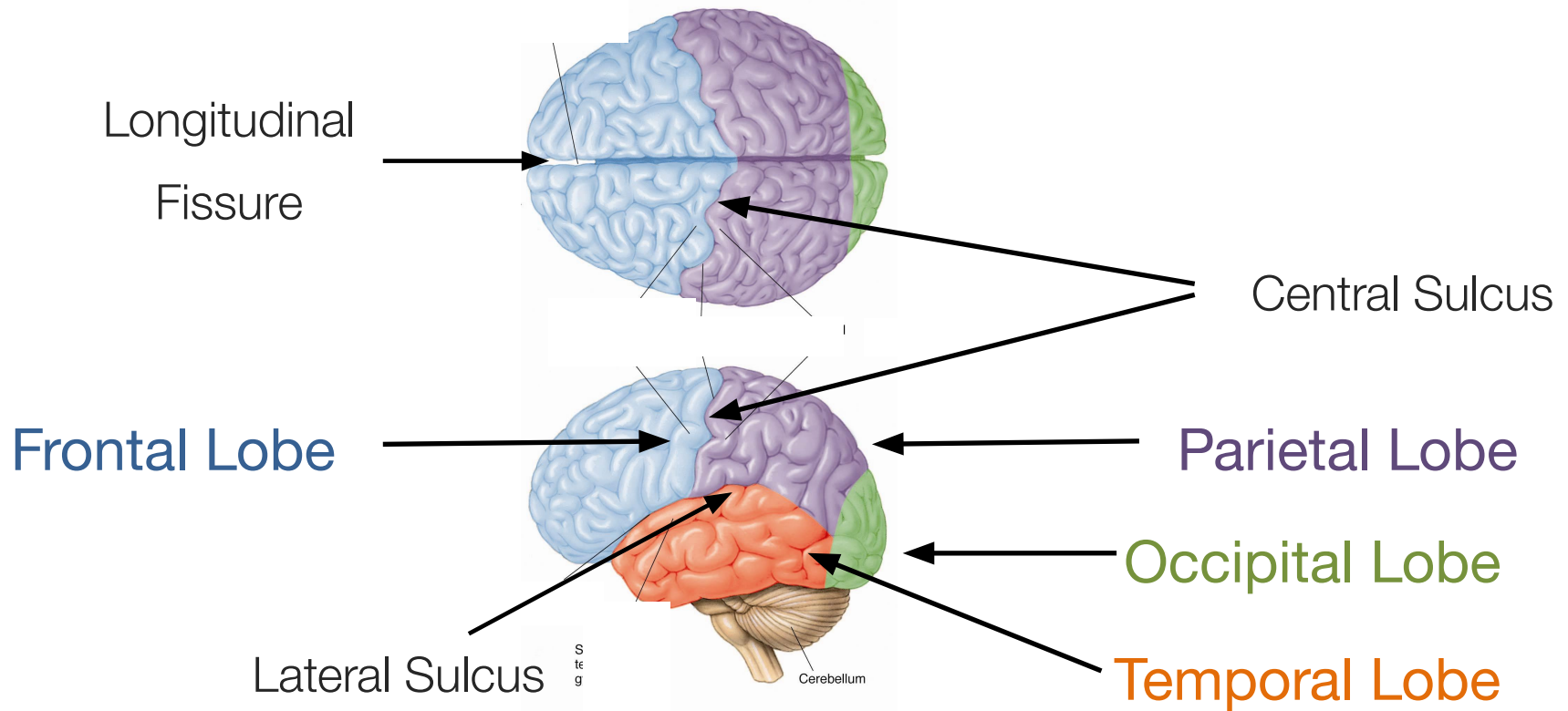


Cerebral cortex

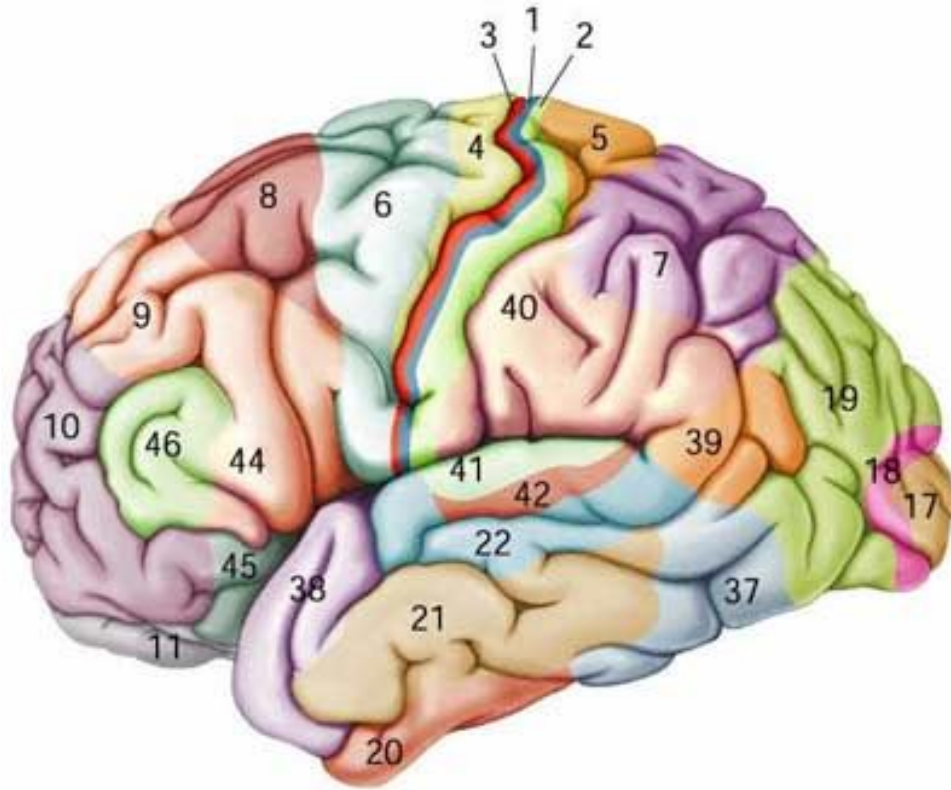
- Cortex (“bark”): outermost layer of the brain
- Deeply folded structure!



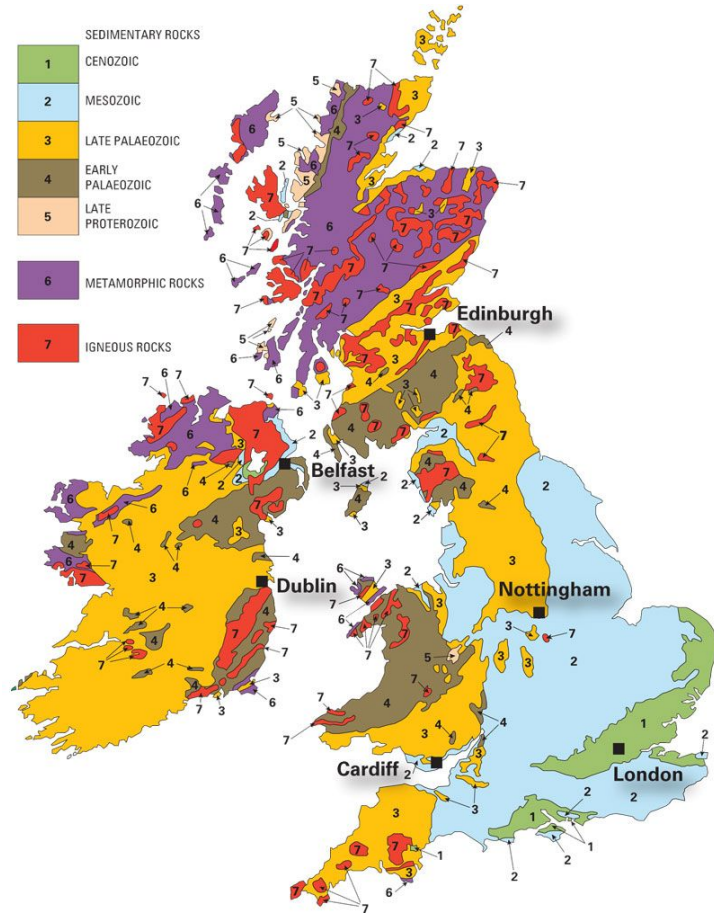
Cerebral cortex



Brodmann's Areas

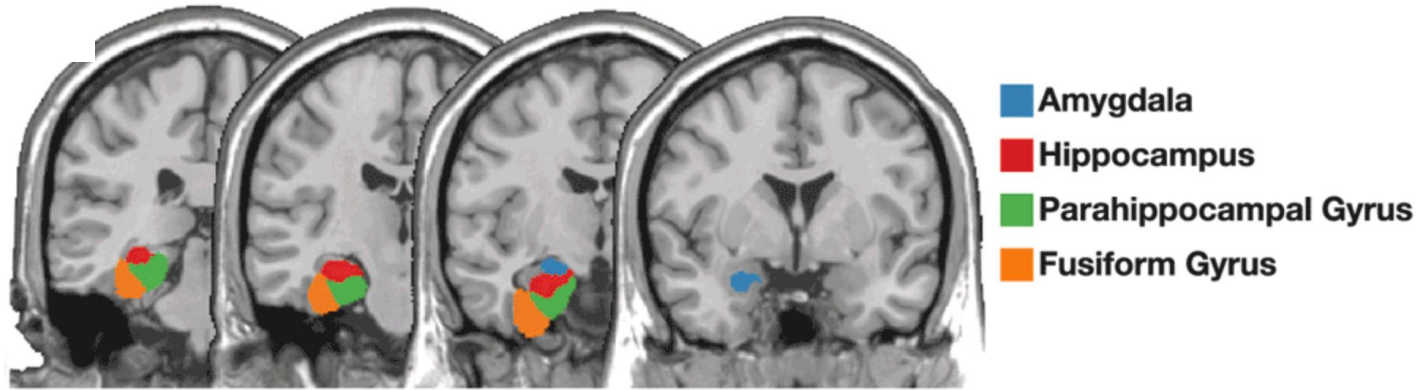


Brodmann's Areas



Brodmann's Areas are almost like a geological map of the brain with divisions reflecting changes in physical properties of the cells.

The Medial Temporal Lobes (MTL)



MTL = hippocampus + parahippocampal gyrus

Parahippocampal gyrus can be decomposed into several smaller regions

Next time...

Working memory

Reading on Canvas